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Nonregular Employment and Payout Policy: Evidence from the Massachusetts Independent Contractor Law

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Abstract. Compared with regular employees, independent contractors (ICs) offer labor flexibility and cost savings to their employers. Using a difference-in-differences design around the 2004 Massachusetts law that discourages IC usage, we find that this exogenous decrease in IC usage makes treated firms' earnings more sensitive to changes in sales, increases labor-related expenses, and reduces profitability. Firms subsequently reduce share repurchases. The decrease is more pronounced for firms with high operating leverage and financial constraints. Our results are robust to entropy balancing. We conclude that IC usage affects firms' operating leverage and profitability, which in turn, influence payout policy.

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Keywords: independent contractor • nonregular employment • repurchases • payout policy • operating leverage

1. Introduction

In the United States, firms either hire permanent employees or utilize temporary contract workers, such as independent contractors (ICs). ICs are self-employed individuals who can offer their services to more than one employer and account for the majority of the workers who are not regular employees.¹ The use of ICs by firms can be beneficial for several reasons. ICs increase a firm's operating flexibility because firms can adjust the number of ICs in response to demand shocks (Abraham and Taylor 1996, Segal and Sullivan 1997, Houseman 2001, Autor 2003, Cappelli and Keller 2013, Ono and Sullivan 2013). ICs can also save costs because they are not subject to the same labor protections as employees, including minimum wage, union rights, and unemployment insurance.² Lim et al. (2019) find that the proportion of ICs in the U.S. workforce increased from 6.6% to 8.1% from 2001 to 2016. Despite the rising popularity of independent contractors, the finance literature offers little evidence on how IC usage affects corporate financial policies.

We examine how constraints on IC usage affect a firm's operations and payout policy. Mauer and Triantis (1994) highlight the trade-off between operating risk and financial risk. From a firm's standpoint, ICs offer more labor flexibility than permanent employees. For example, firing employees with open-ended contracts may lead to legal hurdles and high severance payments, but

companies can dismiss ICs with relatively low costs because ICs are hired on a short-term or project basis. Thus, a higher (lower) proportion of ICs in a firm's workforce implies a more flexible (rigid) labor cost structure, consistent with a lower (higher) degree of operating leverage (or lower (higher) sensitivity of changes in profits to changes in sales). Forcing a firm to lower the proportion of ICs may thus make it more vulnerable to decreases in revenue, and firms may reduce corporate payouts as a precautionary measure.

However, variation in IC usage may not be significant enough to affect a firm's operations given that regular employees constitute the majority of a firm's workforce. For example, the 2005 BLS Contingent and Alternative Employment Arrangement survey finds that there are 10.3 million ICs in the United States, and Lim et al. (2019) estimate that there are 147 million regular employees in the same year. Thus, despite the recent increase in the proportion of ICs in a firm's workforce, it is not ex ante clear whether IC usage is significant enough to affect a firm's operating leverage and payouts.

To address the endogenous relation between IC usage and firm policies, we exploit the 2004 Massachusetts Independent Contractor Law (MICA) that imposes strict requirements for classifying workers as ICs. Because firms do not pay certain labor-related expenses (e.g., unemployment insurance premiums) for ICs, companies have incentives to misclassify workers as independent

contractors. Belman and Block (2008) argue that such misclassification may reduce firms' labor costs by 20%–40%, at the expense of the misclassified workers' welfare. The 2004 MICL makes this misclassification more difficult by imposing harsher penalties, broadening the scope of the statute, and applying a stricter definition of ICs. Consequently, many firms doing business in Massachusetts had to either reclassify some ICs as employees or not renew the contracts of their existing ICs. In either case, the MICL reduces the proportion of ICs in the workforce of Massachusetts firms, making these firms' labor structure more rigid.

Using a difference-in-differences (DID) specification, we estimate the causal effect of ICs on firms' operations and payout policy. We test how operating performance and payout change for treated firms (Massachusetts firms) relative to the control group (non-Massachusetts firms) surrounding the 2004 MICL adoption. We first find that the stricter requirements for classifying ICs raise labor-related expenses and operating costs (cost of goods sold (COGS) plus selling, general, and administrative costs (SG&A)), thus reducing operating profitability.³ The increase in labor expenses is likely driven by increased compensation (e.g., more overtime pay) to the existing workforce, the loss in cost savings associated with using ICs, operating disruptions caused by ICs leaving after their contracts expire, or legal liabilities/settlements because of the misclassified workers.⁴ The increase in SG&A is consistent with a more rigid cost structure because SG&A is not responsive to sales fluctuations (Chen et al. 2019). We also test whether the stricter requirements for using ICs are value relevant for stock market participants. Based on the negative cumulative abnormal returns (CARs) for the treated group (alone and relative to the control group) surrounding the adoption of the 2004 Massachusetts law, we argue that investors view the law unfavorably, consistent with the subsequent decrease in profitability.

We next examine how operating leverage changes after the MICL adoption. Because ICs are more flexible than employees, a decrease in IC usage implies lower labor flexibility. Lower labor flexibility signals that a firm's labor-induced operating leverage is greater (e.g., Serfling 2016). Consistent with such prediction, we find that the treated group's earnings before interest and taxes (EBIT) becomes more sensitive to changes in sales after the MICL adoption. Thus, as costs become more fixed because of the decrease in IC usage, the earnings-sales sensitivity increases.

Given the inverse relation between operating risk and financial risk, we next examine whether the curbed IC usage leads to lower payout.⁵ Although the MICL adoption may affect both dividends and stock repurchases, the effect should be more pronounced for stock repurchases than for dividends. Dividends are "sticky," and firms smooth dividend payments over

time and are reluctant to reduce or omit dividends because doing so sends a negative signal to investors (Lintner 1956, Brav et al. 2005, Leary and Michael 2011). In contrast, stock repurchases are more flexible (Bonaimé et al. 2014), do not necessarily signal permanent payments, and are associated with business cycles or transient cash flows (Guay and Harford 2000, Jagannathan et al. 2000). Thus, if a firm reduces its payout in response to a shock to its ability to use ICs, we expect the effect to be more pronounced for repurchases than for dividends.

We find that the exogenous decrease in IC usage reduces the payouts of Massachusetts firms relative to the control group. This effect is driven by a reduction in stock repurchases. We do not find any significant variation in dividends following the law adoption. The economic magnitude of the effect on repurchases is sizable. Following the regulatory change, firms decrease stock repurchases by 22% relative to the sample mean. In sum, the restrictions on IC usage imposed by the MICL lower profitability and raise a firm's operating leverage, which in turn, reduce payout, consistent with DeAngelo et al. (2004) and Kulchania (2016), respectively.

Next, we examine the conditions under which the impact of the decrease in IC usage on firm outcomes is more pronounced. First, we find that the increase in operating leverage and decrease in corporate payouts are more pronounced for firms that are more impacted by the law (i.e., firms with a higher proportion of ICs in their workforce). Second, we find that financially constrained or distressed firms, which need more internal funds to withstand adverse shocks, are more likely to reduce stock repurchases and total payout. Finally, we find that the decrease in corporate payouts is more pronounced for firms with higher operating leverage.

Lastly, we check whether our inferences remain robust to alternative specifications. Results based on entropy balancing (Hainmueller 2012, Hainmueller and Xu 2013) alleviate the concern that our findings may be driven by the difference in fundamentals between the treated group and the control group. We use a staggered DID design including similar law adoptions in other states (Deknatel and Hoff-Downing 2015) during our sample period and show a comparable treatment effect on a firm's operations and payout policy. In the online appendix, we show that our findings are not driven by firms moving headquarters across states, U.S. multinationals located outside of Massachusetts, and the 2007 minimum wage increase in Massachusetts. Our findings hold when the control variables are omitted to avoid the "bad controls" problem (Angrist and Pischke 2009).

Our study contributes to the literature in two important ways. First, we find novel evidence that a firm's use of independent contractors affects corporate financial policy. The literature on the relation between labor market frictions and financial policy has focused on regular employees (e.g., Matsa 2010, Agrawal and Matsa 2013,

Serfling 2016, Ghaly et al. 2017, Klasa et al. 2018), although there are a few exceptions. Chino (2021) and Kuzmina (2023) argue that nonregular employment affects financial outcomes by changing a firm's operating leverage. They find that a more flexible labor structure reduces the cost of external financing and increases a firm's debt capacity, respectively. Hahn et al. (2021), on the other hand, argue that an exogenous decrease in the use of temporary workers induces a firm to strategically reduce cash holdings to limit labor bargaining power. Our work is consistent with Chino (2021) and Kuzmina (2023) in the sense that the variation in operating leverage caused by the decreased IC usage affects corporate financial outcomes.

Second, we contribute to the literature on the determinants of corporate payouts (e.g., Farre-Mensa et al. 2014). Prior studies offer evidence on how union bargaining power and employee firing costs affect payout policy (e.g., Chino 2016, Dang et al. 2021). We use the 2004 MICL as a quasinatural experiment and find that the law resulted in labor market frictions that decreased the ability of firms to use ICs. Affected firms reduce repurchases as a result.

2. Institutional Background and Hypothesis Development

2.1. Why Do Companies Use Independent Contractors?

In the U.S., firms classify their workers as either employees or independent contractors. The use of independent contractors instead of employees can often provide firms with both flexibility and cost savings. Because contracts with ICs tend to be on a short-term or project basis, firms can adjust the size of their workforce more quickly in response to variation in demand than they could with employees. For example, an accounting firm may hire a self-employed accountant to audit a client or to process tasks during busy seasons. IC usage can also decrease fixed costs in multiple ways. Because contract workers often possess specialized skills or knowledge, they do not require extensive training when hired. Firms are not obliged to withhold Federal Insurance Contribution Act (FICA) and Federal Unemployment Tax Act (FUTA) excise taxes for ICs. Contract workers are also not entitled to minimum wage or overtime pay. Finally, firms do not contribute to state unemployment or workers compensation insurance for ICs and thus, save costs related to payroll tax compliance and worker supervision.

2.2. Independent Contractor Misclassification

To save on labor-related expenses and to circumvent their legal liabilities under various employment laws, employers have incentives to classify employees as ICs. Such practices harm the welfare of misclassified workers, however, if these workers are paid less than minimum wage or deprived of overtime pay and other

employee protections. Misclassification also reduces tax revenues for federal and state governments because firms are not obliged to withhold labor-related taxes for ICs. Carré and Wilson (2004) estimate that from 2001 to 2003, the Massachusetts government lost between \$103 million and \$187 million in annual employee protection-related tax revenue because of IC misclassification. These tax "savings" may reduce firms' labor costs by 20%–40% (Belman and Block 2008).

De Silva et al. (2000) find that approximately 30% of audited firms misclassify their workers. Not every misclassification, however, is intentional. According to Reibstein (2015), most firms misclassify workers because they misunderstand the current regulation. Because federal and state governments apply their own tests to determine a worker's legal status, a firm needs to be aware of both federal regulations and state-level laws. The U.S. Government Accountability Office (2006, p. 2) points out, "the tests used to determine whether a worker is an independent contractor or an employee are complex, subjective, and differ from law to law." This confusion may add to the unintentional misclassification of workers by firms.

Regardless of whether misclassification is intentional or not, the risk can be substantial. Employers who misclassify workers as ICs may be liable for any foregone FICA and FUTA contributions or income tax liability that should have been withheld if the worker had been classified as an employee. If a misclassified worker has been paid less than minimum wage or deprived of overtime pay, the firm is subject to monetary liability for the foregone labor expenses. State governments may also claim a liability for the unpaid state unemployment insurance premiums and worker compensation insurance premiums. An enforcing authority may impose civil or criminal penalties for misclassification, and the misclassified workers may file a lawsuit against their employers.⁶ For example, a cleaning and janitorial services company, CleanNet USA Inc., had to pay \$10 million to settle a class action lawsuit in Massachusetts for misclassifying employees as independent contractor "franchisees" (Reibstein 2013).

2.3. Worker Status: Federal Regulation Vs. the MICL

The Internal Revenue Service (IRS) is responsible for determining whether a worker is an employee or independent contractor for the purpose of federal employment taxes. The principal test applied by the IRS to determine a worker's status is called the "common law test." This test determines worker classification based on a firm's right to control a worker in terms of where, when, and how the work is performed. The IRS considers 20 factors in applying the common law test.⁷ U.S. businesses over time have become familiar with the IRS test in worker classification. Thus, the 20-factor test is the standard for classifying workers. However, the situation

gets more complicated, especially for firms that operate in multiple states because different states have their own tests for determining worker status.

Massachusetts uses a three-part test (the “ABC test”) for determining worker status for unemployment compensation purposes. According to the ABC test, if an employer argues that a worker is an independent contractor, it must establish

(a) such individual has been and will continue to be free from control and direction in connection with the performance of such services, both under his contract for the performance of service and in fact; and

(b) such service is performed either outside the usual course of the business for which the service is performed or is performed outside of all the places of business of the enterprise for which the service is performed; and

(c) such individual is customarily engaged in an independently established trade, occupation, profession or business or the same nature as that involved in the service performed.

In 1990, Massachusetts enacted the Independent Contractor Law (“the MICL”). The MICL followed the ABC test, which previously pertained only to unemployment compensation law. Massachusetts firms, especially non-construction businesses, were largely unaware of the 1990 MICL, however, and continued to use the IRS common law test to determine worker status. In 2004, the MICL was amended, and its provisions became effective immediately. This amendment expanded the application of the original MICL from unemployment compensation purposes to wage law requirements, including minimum wage and overtime pay.⁸ The amendment also increased the monetary penalty for both intentional and unintentional misclassification. Arguably, the most crucial part of the 2004 revision is that it expanded the applicability of the statute by amending the ABC test in a significant way; it deletes the “or is performed outside of all places of business of the enterprise for which the service is performed” phrase from the second provision of the ABC test. Prior to this revision, a Massachusetts firm could classify a worker as an IC as long as the worker remained in a location outside any establishment of a firm, even if the worker was performing tasks related to the firm’s usual course of business. From 2004 onward, any worker in charge of tasks related to a firm’s day-to-day operations must be classified as an employee, regardless of the worker’s location. Thus, any business that hired ICs who worked at home or at their own business office was impacted. As a result, in December 2004, the Massachusetts Attorney General issued an advisory declaring that the newly amended statute “excludes far more workers from independent contractor status than are disqualified under the IRS (Internal Revenue Service) common law test.”⁹ Arguably, the Massachusetts test is the toughest in the country for IC classification.¹⁰ Unlike the 1990 MICL, the 2004 amendments received

significant attention. According to Shea (2005), however, most legislators did not anticipate the repercussions of the MICL, especially on nonconstruction firms, and business groups outside the construction industry were unaware of the legislation before it passed. Thus, the MICL is plausibly an “exogenous” shock to Massachusetts firms.

In summary, the 2004 MICL offers a much stricter worker classification test than the IRS test. Thus, after the passage of the MICL, Massachusetts businesses had to thoroughly re-examine worker status to ensure they are free from any liability because of misclassification, regardless of whether the misclassification has been intentional or not. Moreover, with these stricter regulations, employees may file lawsuits against their employers for misclassification, and the resulting costs imposed on an employer may be substantial.

2.4. Hypothesis Development

Following the 2004 MICL amendment, many firms doing business in Massachusetts had to either reclassify some of their former ICs as employees or not renew the employment contracts of ICs who do not meet the definition according to the 2004 MICL. In either case, the regime shift reduces the proportion of ICs in Massachusetts firms, making these firms’ labor structure more rigid. If firms reclassify, then they incur additional expenses for the reclassified workers, including employee benefits, state unemployment insurance and worker compensation premiums, and withholding federal taxes for the reclassified workers (Wood 2009). These expenses increase fixed labor costs and likely lead to higher labor-induced operating leverage. If firms choose to not renew the contracts, then the existing employees must do the same amount of work with fewer people. These workers may demand higher compensation for this heavier workload. Regardless of whether a firm reclassifies workers as employees or does not renew IC contracts, a company likely faces lower operating flexibility because of reduced IC usage following the law adoption.¹¹ Consistent with the MICL making it difficult for Massachusetts firms to use ICs, Lim et al. (2019) find that Massachusetts ranks 33rd among the 50 U.S. states plus the District of Columbia in terms of IC growth from 2001 to 2016.

Our hypothesis is based on the idea that the 2004 MICL affects corporate payout policy because the revision curbs the use of ICs, makes a firm’s labor structure more rigid, and increases a firm’s operating leverage. When facing shortfalls in cash flow, firms want to lay off workers (Ofek 1993, Kang and Shivdasani 1997). Terminating the contracts of ICs is simple because IC contracts are normally short term, and firms can let them expire. However, regular employees have open-ended contracts, and firing them may involve legal hurdles and significant severance payments. Thus, reducing the proportion of ICs in a firm’s workforce makes the firm less flexible. Serfling (2016)

finds that labor inflexibility for regular employees increases a firm's operating leverage, and Kulchania (2016) argues that firms with higher fixed costs reduce corporate payouts. The higher labor rigidity caused by the MICL may increase labor adjustment costs and a firm's liquidity problem. Also, the exogenous decrease in IC usage implies a loss in cost savings associated with employing nonregular workers. Thus, to the extent that MICL lowers the proportion of ICs and reduces operating flexibility and profitability, firms reduce payouts, especially more flexible stock repurchases.

3. Data and Empirical Methods

3.1. Sample Selection

Our sample covers publicly held Compustat industrial firms (excluding financials and utilities) from 2000 to 2007, which are the eight years surrounding the 2004 MICL adoption. Following Farre-Mensa et al. (2014), we require firms to be in both Compustat and Center for Research in Security Prices (CRSP); be incorporated in the United States; be listed on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), or the Nasdaq Stock Market (NASDAQ); have valid stock prices in CRSP; and have a CRSP share code of 10 or 11. We exclude observations with major merger events (Compustat footnote code "AB") and only consider firm-years with nonmissing, positive values for book assets and sales. We set repurchases to zero if missing. We require observations to have nonmissing values for the firm-level variables, such as profitability, dividends, capital expenditure, retained earnings, market-to-book, leverage, cash holdings, size, age, systematic and idiosyncratic risk, and the state-level variables (Gross Domestic Product (GDP) growth rate and population). To adjust for inflation, we examine real dollar amounts using the price level in 2009.

We obtain historical headquarter addresses using the "Augmented 10-X Header Data" presented by the Software Repository for Accounting and Finance.¹² Following Smith (2016), we supplement missing historical addresses using the Compustat header file. Similar to previous studies (e.g., Agrawal and Matsa 2013, Serfling 2016, Dang et al. 2021), we assume that the majority of a firm's workers, including independent contractors, are situated in the state where the firm is headquartered.¹³ The final sample has 29,157 firm-year observations.

3.2. Empirical Models

To study how decreased usage of ICs affects corporate policies, we construct a DID setting in which we compare changes in the variables of interest between "treated" and "control" firms surrounding the MICL adoption. Because the 2004 MICL covers all industries, we compare Massachusetts firms (treated) with non-Massachusetts firms (control) using the following empirical specification:¹⁴

$$DepVar_{it} = b_1 Treat \times Post_{st} + \gamma' X_{ist} + \tau_s + v_i + w_{jt} + e_{it}, \quad (1)$$

where i , s , j , and t denote firm, firm's headquarter state,

industry, and year, respectively. $DepVar$ is the annual firm-level operating performance or payout variable. For operating performance, we focus on SG&A and operating income before depreciation and amortization scaled by book asset, respectively. Because of a lack of data on firm-level staff expense (Compustat XLR), we use SG&A as a proxy for labor-related expenses (e.g., Lee et al. 2018).¹⁵ We use three payout measures as a dependent variable: cash dividends, the dollar value of share repurchases, and the sum of dividend payments and repurchases; all are scaled by book assets. $Treat$ is an indicator variable equal to one if a firm is headquartered in Massachusetts and zero otherwise. $Post$ is an indicator variable equal to one if a firm's fiscal year end comes in June 2004 or afterward and zero otherwise.¹⁶ $Treat \times Post$ shows how operating performance or payout policy changes for the treated group relative to the control group after the regulatory shift in Massachusetts.

We include firm fixed effects to control for time-invariant firm characteristics, industry (two-digit Standard Industrial Classification (SIC) code)-by-year fixed effects to account for unobserved, time-varying characteristics at the industry level, and headquarter state fixed effects to reflect unobserved, time-invariant state-level characteristics.¹⁷ Controls include cash holdings, capital expenditure, market-to-book ratio, financial leverage, firm size (NYSE market-cap percentile), the natural log of a firm's age, idiosyncratic risk, state GDP growth rate, and the natural log of state population in the baseline regression.¹⁸ Appendix A presents variable definitions. We adjust standard errors for heteroscedasticity and cluster them at the state level to correct for serial correlation within a state (Bertrand et al. 2004).

3.3. Distribution of Independent Contractors and Descriptive Statistics

Table 1 shows the distribution of independent contractors across the U.S. industries. We rely on the 2005 BLS Contingent and Alternative Employment Arrangements survey. The data are collected in supplements to the Current Population Survey, which is a nationwide periodic survey of U.S. households that gathers information on employment and other demographic characteristics. The 2005 survey provides the distribution of independent contractors across 14 industries classified by the BLS according to North American Industry Classification Systems (NAICS).¹⁹ Twenty-two percent of the ICs in the United States are construction workers, and 21.3% of them are in "professional and business activities." "Mining" and "public administration" only account for 0.1% and 0.3% of the ICs, respectively. Independent contractors are not necessarily concentrated in either high-tech or low-tech industries. We exclude "financial activities" and "transportation and utilities" from consideration because we do not cover financials and utilities in this paper. We also omit "public administration"

Table 1. Distribution of Independent Contractors Across Industries

Industry (NAICS code)	Percentage
Construction (23)	22.0
Professional and business activities (54–56)	21.3
Financial activities (52 and 53)	10.4
Other services (81)	9.9
Retail trade (44 and 45)	8.9
Education and health services (61 and 62)	8.7
Leisure and hospitality (71 and 72)	4.5
Transportation and utilities (48 and 49, 22)	3.9
Manufacturing (31–33)	3.2
Agricultural and related industries (11)	2.6
Wholesale trade (42)	2.1
Information (51)	2.0
Public administration (92)	0.3
Mining (21)	0.1

Notes. This table shows the distribution of independent contractors across the 14 industries in the United States. The information is from the 2005 Contingent and Alternative Employment Arrangements survey by the BLS. We exclude “financial activities” and “transportation and utilities” from consideration because we focus on industrial firms, and we exclude “public administration” because there is zero observation in our sample after initial screening. The BLS survey does not present a more detailed distribution.

from consideration because we have zero observations in our full sample.

Table 2 reports the descriptive statistics for the variables in this paper. The recurring operating costs, profits, the payout measures, and the firm-level controls are winsorized at the 1st and 99th percentiles.²⁰ On average, stock repurchases amount to 1.78% of its book assets, whereas dividends only account for 0.55% of a firm’s book assets. These statistics show that stock repurchases

are more pronounced than dividends during the sample period, consistent with Skinner (2008). About 5.9% of the observations in the full sample belong to the treatment group, and the observations after the law adoption cover 43% of the full sample.

4. Results

4.1. Operating Expenses, Stock Market Reactions, and Operating Leverage

The major advantages of using ICs are labor flexibility and cost savings. Consequently, we first ascertain that the exogenous decrease in IC usage affects a firm’s cost structure. In particular, a firm’s labor cost structure should be more rigid after the law adoption because it is more difficult to lay off regular employees than to terminate the contract of ICs. Thus, a lower proportion of ICs implies higher fixed labor costs for a firm, which implies higher operating leverage. Moreover, labor-related expenses may increase. If the treated firms have misclassified some of their workers and the 2004 MICL adoption imposes strict criteria for using ICs, then the proportion of ICs in a firm’s workforce should decrease. If firms reclassify ICs as employees, then the companies pay additional fixed payments related to employee benefits (e.g., unemployment insurance premium). If firms do not reclassify and let the employment contract of ICs expire, then the companies end up with a smaller workforce.²¹ The remaining workers can then demand more compensation for their additional workload. The MICL may also lead to operating disruptions because the company loses ICs who have experience in processing certain tasks for the firm. The cost savings associated with using

Table 2. Summary Statistics

Variable	N	Mean	Median	Standard deviation	25th percentile	75th percentile
<i>Dividends/Assets</i>	29,157	0.0055	0.0000	0.0139	0.0000	0.0025
<i>Repurchase/Assets</i>	29,157	0.0178	0.0000	0.0473	0.0000	0.0082
<i>Total Payout/Assets</i>	29,157	0.0251	0.0001	0.0581	0.0000	0.0208
<i>Labor Expenses/Assets</i>	29,157	0.3244	0.2497	0.3039	0.1103	0.4438
<i>Operating Costs</i>	29,157	1.0857	0.8962	0.7995	0.5293	1.4037
<i>Treat</i>	29,157	0.0587	0.0000	0.2351	0.0000	0.0000
<i>Post</i>	29,157	0.4308	0.0000	0.4952	0.0000	1.0000
<i>Cashflow (or Profitability)</i>	29,157	0.0166	0.0942	0.2763	−0.0122	0.1572
<i>Cash_Hold</i>	29,157	0.2309	0.1276	0.2506	0.0324	0.3619
<i>Capex</i>	29,157	0.0527	0.0327	0.0601	0.0164	0.0642
<i>Market-to-Book</i>	29,157	2.1766	1.5691	1.8301	1.1322	2.4612
<i>Debt</i>	29,157	0.2017	0.1451	0.2179	0.0058	0.3230
<i>Size</i>	29,157	0.3277	0.2529	0.2904	0.0600	0.5412
<i>log_Age</i>	29,157	2.3228	2.3979	0.9682	1.7918	2.9957
<i>Idiosyncratic Volatility</i>	29,157	0.0398	0.0330	0.0242	0.0220	0.0513
<i>StateGDP_Growth</i>	29,157	0.0532	0.0509	0.0243	0.0373	0.0687
<i>log_StatePopulation</i>	29,157	16.2411	16.2579	0.8502	15.6058	16.9080

Notes. This table presents descriptive statistics for the variables in our baseline regression. The sample period spans from 2000 to 2007. The sample contains 29,157 firm-year observations. We only include firms with positive, nonmissing assets, sales, and industrial classification, and we exclude financials (SIC 60–69) and utilities (SIC 49). We obtain a firm’s historical address each year from Augmented 10-X Header Data provided by the Software Repository for Accounting and Finance at the University of Notre Dame. We supplement the historical addresses with Compustat header addresses (Smith 2016). The variable definitions are in Appendix A.

ICs are lost or at least limited. The treated firms may also be subject to legal liabilities or settlements²² because of the unpaid employee protection-related taxes or a lawsuit by the misclassified workers.²³

In Table 3, we test how labor-related expenses (proxied by SG&A), operating costs, and profitability change in response to the decrease in IC usage. Column (1) shows that labor-related expenses increase for the treated group relative to the control by 2.3% relative to book assets or 7.1% relative to the sample mean after the regulatory shift. The loss in labor flexibility and cost savings associated with ICs seems to induce significant cost increases for the affected firms. Column (2) shows that the increase in total operating costs (COGS plus SG&A) is also statistically significant. In other words, the increase in labor-related expenses for the treated firms is large enough to affect their total operating costs. The increase in SG&A is also consistent with a more rigid cost structure because SG&A is not responsive to sales

fluctuations (Chen et al. 2019). Column (3) confirms that the cost increase for the treated firms following the MICL adoption reduces their operating profitability by 1.5% relative to book assets. In sum, Table 3 corroborates that the exogenous decrease in IC usage, caused by the MICL adoption, increases labor-related expenditures, which in turn, decreases profitability.

Next, we examine how stock market participants reacted to the MICL adoption. Given the negative effect of the regulation on profitability, we can expect the stock price reactions of the treated firms to be negative when the law became binding. Although the MICL was signed into law on July 19, 2004, the related enforcement statute was amended on June 10, 2004 (Deknatel and Hoff-Downing 2015). Because the earlier revision of the state statute was the first news available to Massachusetts businesses regarding independent contractor classification, we use June 10, 2004 as the event date of interest ($t = 0$). Stock returns are measured by CARs over the three

Table 3. Labor-Related Expenses, Operating Costs, and Profitability

	(1) <i>Labor Expenses over Assets</i>	(2) <i>Operating Costs</i>	(3) <i>Profitability</i>
<i>Treat × Post</i>	0.0229*** (3.638)	0.0317** (2.286)	−0.0147*** (−2.868)
<i>Cash_Hold</i>	−0.1473*** (−8.507)	−0.5396*** (−12.967)	0.0196 (1.341)
<i>Capex</i>	0.4044*** (4.070)	0.7479*** (4.846)	−0.2838** (−2.490)
<i>Market-to-Book</i>	0.0294*** (16.074)	0.0808*** (19.186)	−0.0133*** (−6.095)
<i>Debt</i>	−0.0402** (−2.342)	−0.1033** (−2.679)	−0.0991*** (−7.386)
<i>Size</i>	−0.5177*** (−9.990)	−1.2008*** (−14.488)	0.4401*** (10.240)
<i>log_Age</i>	0.0101* (1.747)	0.0147 (1.258)	0.0181*** (2.907)
<i>Idiosyncratic Volatility</i>	0.9048*** (5.585)	2.2501*** (8.438)	−1.0954*** (−4.649)
<i>StateGDP_Growth</i>	0.0724 (1.116)	0.1408 (1.067)	−0.0044 (−0.068)
<i>log_StatePopulation</i>	0.1483 (1.384)	0.3319 (1.237)	−0.0385 (−0.508)
Headquarter state FE	Yes	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	29,157	29,157	29,157
Adjusted R^2	0.801	0.871	0.794

Notes. This table shows how curbing the use of independent contractors affects a firm's labor-related expenses, operating costs, and profitability. Because of a lack of data on staff expenses (Compustat XLR), we use SG&A as a proxy for labor expense (e.g., Lee et al. 2018). We set SG&A to zero if missing. For the dependent variables, we use SG&A, operating costs (COGS plus SG&A), and operating profit before depreciation and amortization scaled by book assets, respectively, for a firm in year t . We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

Table 4. CARs Around the Adoption of the Massachusetts Independent Contractor Law

CAR window	Obs. = 199		
	(1) Market model CARs	(2) 3-factor CARs	(3) 4-factor CARs
[−13, −4]	−0.41% (−0.69)	−0.54% (−0.91)	−0.54% (−0.89)
[−3, −2]	−1.53%*** (−4.49)	−1.21%*** (−3.58)	−1.21%*** (−3.58)
[−1, +1]	−2.27%*** (−5.61)	−1.26%*** (−3.17)	−1.29%*** (−3.19)

Notes. This table shows stock price reactions to the adoption of the Massachusetts Independent Contractor Law. The event date of interest ($t = 0$) is June 10, 2004 when the enforcement statute was revised (Deknatel and Hoff-Downing 2015). We cover firms headquartered in Massachusetts in 2004 with nonmissing CRSP PERMNOs. Financial and utilities are excluded. We use the Event Study by WRDS tool to compute CARs. Columns (1), (2), and (3) are based on the market model, the Fama and French (1993) three-factor model, and the Carhart (1997) four-factor model, respectively. We require an estimation window of 252 days and a 30-day gap. Each firm should have at least 100 days with nonmissing returns within the estimation window. CARs are computed over the pre-event window (trading days [−13, −4] and [−3, −2]) and event window (trading days [−1, +1]). The t statistics are presented in parentheses. Appendix A presents the variable definitions.

***Statistical significance at the 1% level.

days surrounding this date (trading days [−1, +1]). We also estimate CARs for the preannouncement period (trading days [−13, −4] and [−3, −2]) to test whether investors anticipated the adoption of the MICL. CARs are based on the market model, the Fama and French (1993) three-factor model, and the Carhart (1997) four-factor model, respectively.

Table 4 shows that CARs are negative but not statistically significant over the [−13, −4] period, regardless of the model used. CARs over trading days [−3, −2] are negative and statistically significant, ranging from −1.21% to −1.53%. In the event window (trading days [−1, +1]), CARs are negative and statistically significant at the 1% level, ranging from −1.26% to −2.27%. Table 4 suggests that the adoption of the MICL was not widely anticipated by investors, although CARs during the trading days [−3, −2] imply some degree of information leakage just prior to the event. Such a result aligns with the anecdotal evidence that “most business groups outside the construction industry were unaware of the legislation before it passed and never weighed in on it.”²⁴ Firms in the control group also show negative stock market reactions ranging from −0.40% to −1.27% in the same window. However, the difference in stock market reactions between the treated group and the control group is significant at the 5% level. Consistent with the decrease in profitability (Table 3), the stock market reactions to the MICL adoption were also negative.

Next, we ascertain that the increased cost rigidity following the decreased IC usage increases a firm’s labor-induced operating leverage. Serfling (2016) shows that higher labor rigidity increases a firm’s operating leverage. If the 2004 revision discourages IC usage and reduces labor flexibility, the sensitivity of changes in EBIT to changes in sales should increase for the treated firms after the MICL adoption.²⁵ We follow Eisfeldt and Papanikolaou (2013) and run the following regression to examine the variation in a firm’s operating leverage

following the 2004 MICL adoption:

$$\begin{aligned} \Delta \log(\text{EBIT})_{it} = & b_1 \text{Treat} \times \text{Post} \times \Delta \log(\text{Sales})_{it} \\ & + b_2 \Delta \log(\text{Sales})_{it} + b_3 \text{Treat} \times \text{Post}_{st} \\ & + \gamma' X_{it} + \mu_s + v_i + w_{jt} + e_{it}, \end{aligned} \quad (2)$$

where i , s , j , and t denote firm, firm’s headquarter state, industry, and year, respectively. We include the firm- and the state-level controls, firm fixed effects, industry-by-year fixed effects, and headquarter state fixed effects. The dependent variable is the change in the natural log of EBIT. The main independent variable of interest is $\text{Treat} \times \text{Post} \times \Delta \log(\text{Sales})$. The coefficient of $\Delta \log(\text{Sales})$ alone shows the sensitivity of changes in EBIT to changes in sales ignoring the effect of the 2004 MICL. The coefficient of $\text{Treat} \times \text{Post} \times \Delta \log(\text{Sales})$ shows how this sensitivity changes in response to the curbed use of ICs for the treated firms.

In column (2) of Table 5, the coefficient of $\Delta \log(\text{Sales})$ shows that, without the effect of the 2004 MICL adoption, a 1% decrease in sales is associated with a 1.618% decrease in EBIT. Following the MICL adoption, however, a 1% decrease in a treated firm’s sales is associated with a 2.178% decline in EBIT, an increase in magnitude of 34.6% ($=0.5595/1.6180$). Thus, Table 5 confirms that the exogenous decrease in IC usage increases a firm’s labor-induced operating leverage.

4.2. Payout Policy

In this section, we examine whether the exogenous decrease in IC usage reduces corporate payouts. Because the MICL adoption makes a firm’s cost structure more rigid, the regulatory shift should reduce the amount of dividends or stock repurchases (Kulchania 2016). Table 6 is based on Equation (1) for the test design. The dependent variables are total payout, dividends, and share repurchases scaled by book assets. In column (1), total payout decreases after the MICL goes into effect. Firms

Table 5. Operating Leverage

	(1) $\Delta \log (EBIT)$	(2) $\Delta \log (EBIT)$
$Treat \times Post \times \Delta \log (Sales)$	0.5658*** (8.153)	0.5595*** (7.872)
$\Delta \log (Sales)$	1.6159*** (22.452)	1.6180*** (22.238)
$Treat \times Post$	−0.0070 (−0.325)	−0.0071 (−0.308)
$Cash_Hold$	0.5759*** (6.392)	0.5766*** (6.558)
$Capex$	−1.2977*** (−6.042)	−1.3038*** (−6.280)
$Market\text{-}to\text{-}Book$	0.0646*** (3.843)	0.0636*** (3.830)
$Debt$	−0.1884** (−2.043)	−0.1760* (−1.877)
$Size$	0.4137*** (5.266)	0.4321*** (5.518)
$\log_Age$	−0.0385 (−1.126)	−0.0418 (−1.142)
$Idiosyncratic\ Volatility$	5.5666*** (5.163)	5.5807*** (5.302)
$StateGDP_Growth$	0.7313* (1.799)	0.8307** (2.028)
$\log_StatePopulation$	0.0134 (0.369)	0.0101 (0.049)
Headquarter state FE	No	Yes
Industry-by-year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	16,939	16,939
Adjusted R^2	0.285	0.285

Notes. This table shows how the exogenous decrease in the use of independent contractors affects firm-level operating leverage. We follow Eisfeldt and Papanikolaou (2013) to examine the variation in operating leverage. $\Delta \log (EBIT)$ and $\Delta \log (Sales)$ are the annual change in the natural log of earnings before interest and taxes and before sales, respectively, for each firm in year t . We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

become more conservative in their payout policy in response to the exogenous decrease in IC usage. The economic magnitude is also sizable. Total payout decreases by 18% ($= -0.0044/0.0251$) relative to the sample mean. However, in column (2), dividends show no significant variation. Firms seem reluctant to reduce cash dividends even though MCL curbs IC usage, consistent with the sticky nature of dividends (Lintner 1956, Jagannathan et al. 2000, Brav et al. 2005, Leary and Michaely 2011). In column (3), stock repurchases decrease by 22% ($= -0.0040/0.0178$) relative to the sample mean. Instead of reducing dividends and sending a negative signal

to capital market participants, firms seem to adjust stock repurchases in response to the decrease in IC usage. The reduction in stock repurchases drives the overall decrease in corporate payouts after the MCL goes into effect. In sum, Table 6 implies that the decrease in IC usage caused by the 2004 MCL adoption is significant for a firm to reduce the amount of payout.²⁶ This decline in payouts seems to be driven by stock repurchases, which are more flexible than dividends.²⁷

4.3. Channel Tests

In this section, we examine the conditions under which our results become more pronounced and whether such channel tests corroborate the idea that higher labor-induced operating leverage following the MCL adoption reduces corporate payouts.

First, we test whether the increase in operating leverage and the decrease in corporate payouts are more pronounced for industries that utilize ICs more intensively. In panel A of Table 7, we use Equation (2) and conduct a subsample analysis based on an indicator corresponding to industries with intensive IC usage. $High_IC$ is equal to one if a firm in year t is in one of the five industries (“construction,” “professional and business services,” “other services,” “retail trade,” and “education and health services”) with relatively intensive IC usage according to the 2005 BLS survey and zero otherwise. We split the full sample based on $High_IC$ and estimate Equation (2). The main independent variable of interest is $Treat \times Post \times \Delta \log (Sales)$, which shows how the treated firms’ operating leverage changes in response to the MCL adoption. We compare the coefficient of $Treat \times Post \times \Delta \log (Sales)$ and see which subsample shows more pronounced results.

In both columns (1) and (2) in panel A, the coefficient of $Treat \times Post \times \Delta \log (Sales)$ is positive and significant, but the relation is more pronounced for the “ $High_IC = 1$ ” subsample in column (1). Firms with more intensive IC usage (“ $High_IC = 1$ ”) show a 2.682% ($= 1.3092 + 1.3725$) decline in EBIT for a 1% decrease in sales, whereas companies with less intensive IC usage (“ $High_IC = 0$ ”) exhibit a 2.119% ($= 0.4432 + 1.6756$) decline in EBIT for a 1% decrease in sales. The difference in the coefficients of $Treat \times Post \times \Delta \log (Sales)$ is statistically significant at the 1% level between the two subsamples ($p = 0.0001$). Consistent with our hypothesis, panel A of Table 7 confirms that the increase in operating leverage is more pronounced for firms with intensive IC usage.

In panel B, we interact $Treat \times Post$ with $High_IC$ and use it as a main independent variable to see whether the decrease in corporate payouts is more pronounced in treated firms with intensive IC usage. The coefficient of $Treat \times Post \times High_IC$ is negative and significant for both stock repurchases and total payout. Following the MCL adoption, firms with a greater concentration of ICs react more strongly to the regulatory shift by reducing payouts more than their counterparts.

Table 6. Independent Contractor Misclassification Statute and Corporate Payout

	(1) Total Payout/Assets	(2) Dividends/Assets	(3) Repurchases Assets
<i>Treat × Post</i>	−0.0044** (−2.596)	−0.0003 (−0.795)	−0.0040** (−2.126)
<i>Cash_Hold</i>	0.0044 (0.774)	0.0016* (2.008)	0.0017 (0.355)
<i>Capex</i>	0.0269 (1.462)	0.0046** (2.088)	0.0192 (1.451)
<i>Market-to-Book</i>	0.0018** (2.278)	0.0005*** (4.375)	0.0007 (1.041)
<i>Debt</i>	0.0058 (1.260)	−0.0028*** (−4.245)	0.0054 (1.376)
<i>Size</i>	−0.0064 (−1.159)	−0.0030*** (−2.695)	0.0024 (0.584)
<i>log_Age</i>	0.0032** (2.045)	−0.0017*** (−3.658)	0.0075*** (4.628)
<i>Idiosyncratic Volatility</i>	−0.0162 (−0.528)	−0.0032 (−0.501)	−0.0183 (−0.879)
<i>StateGDP_Growth</i>	−0.0252 (−1.109)	0.0015 (0.407)	−0.0260 (−1.403)
<i>log_StatePopulation</i>	0.0208 (0.661)	0.0124 (1.247)	−0.0040 (−0.114)
Headquarter state FE	Yes	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	29,157	29,157	29,157
Adjusted R^2	0.309	0.597	0.279

Notes. This table reports how curbing the use of independent contractors among a firm's workforce affects corporate payout policy. We take total payout, dividends, and share repurchases scaled by book assets as the dependent variables. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

Second, we test whether the decrease in corporate payouts is more pronounced for financially constrained firms or those in financial distress. Such firms may have difficulty coping with increased operating leverage because of limited access to external financing or lack short-term liquidity or long-term solvency. Consequently, financially constrained or distressed firms may reduce payouts to obtain funds to cover the increased operating costs (Table 3). Thus, if the exogenous decrease in IC usage implies higher operating risk and reduces labor flexibility, then the effect of the MICL adoption should be more pronounced for firms with financial constraints or distress.

In Table 8, we create indicators for financially constrained or distressed firms following Whited and Wu (2006), Hadlock and Pierce (2010), and the Altman Z score. Hadlock and Pierce (2010) create a measure called the Size and Age Index, arguing that only firm size and age predict firm-level financial constraints. Larger and more mature firms tend to be less constrained than their counterparts. Whited and Wu (2006) use generalized method of moments estimation of an investment Euler

equation to create their index. The Whited and Wu (2006) index implies that financially unconstrained firms are large and profitable, have low leverage, are in low-growth industries, and pay dividends. A higher value of the Size and Age Index and the Whited and Wu index implies more binding financial constraints for a firm. The Altman Z score is based on firm-level profitability, leverage, liquidity, long-term solvency, and business activity. The score evaluates a firm's likelihood of bankruptcy, and a lower Z score implies that a company is closer to liquidation.

Based on these measures, we create three indicators for financially constrained or distressed firms. *High HP* is an indicator variable equal to one if a firm in year t has an above-median value of the Size and Age Index for the full sample and zero otherwise. *High WW* is an indicator equal to one if a firm in year t has an above-median value of the Whited and Wu (2006) index for the full sample and zero otherwise. *Low Z* is an indicator equal to one if a firm in year t has a below-median value of Altman Z score for the full sample and zero otherwise. We interact each indicator

Table 7. Intensive Usage of Independent Contractors

Panel A: Operating leverage		
	<i>High_IC</i> = 1 (1) $\Delta \log(\text{EBIT})$	<i>High_IC</i> = 0 (2) $\Delta \log(\text{EBIT})$
<i>Treat</i> $\times$ <i>Post</i> $\times$ $\Delta \log(\text{Sales})$	1.3092*** (5.753)	0.4432*** (5.386)
	$H_0: \beta(1) = \beta(2)$ ($p = 0.0001$)	
$\Delta \log(\text{Sales})$	1.3725*** (8.590)	1.6756*** (24.026)
<i>Treat</i> $\times$ <i>Post</i>	0.0454 (1.024)	−0.0391 (−1.491)
<i>Cash_Hold</i>	0.7091*** (2.983)	0.5131*** (6.620)
<i>Capex</i>	−2.0882*** (−3.510)	−1.1245*** (−6.098)
<i>Market-to-Book</i>	0.0759*** (3.623)	0.0589*** (3.608)
<i>Debt</i>	−0.0343 (−0.245)	−0.2278** (−2.606)
<i>Size</i>	0.6022*** (3.137)	0.3812*** (3.587)
$\log_Age$	−0.0110 (−0.143)	−0.0510 (−0.915)
<i>Idiosyncratic Volatility</i>	2.6644 (1.155)	6.4047*** (5.129)
<i>StateGDP_Growth</i>	0.1702 (0.200)	1.0291* (1.981)
$\log_StatePopulation$	1.2026* (1.860)	−0.3714* (−1.788)
Headquarter state FE	Yes	Yes
Industry-by-year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	3,821	13,118
Adjusted R^2	0.220	0.303

with *Treat* $\times$ *Post* to test whether the decrease in stock repurchases and total payout is more pronounced for financially constrained or distressed firms.

Columns (1) and (2) use *High HP* to measure constraints. The results show a more pronounced decline in stock repurchases and total payout for financially constrained firms. The coefficient of *Treat* $\times$ *Post* $\times$ *High HP* is negative and statistically significant at the 1% level for both columns. Columns (3) and (4) measure constraints using the Whited and Wu (2006) index. Similar to columns (1) and (2), the coefficient of *Treat* $\times$ *Post* $\times$ *High WW* is negative and significant at the 1% level, whereas the coefficient of *Treat* $\times$ *Post* alone is statistically insignificant. Thus, columns (1)–(4) show that financially constrained firms drive the decrease in corporate payouts after the exogenous decrease in the use of ICs. Columns (5) and (6) also show that the decrease

Table 7. (Continued)

Panel B: Payout policy		
	(1) <i>Repurchases/Assets</i>	(2) <i>Total Payout/Assets</i>
<i>Treat</i> $\times$ <i>Post</i> $\times$ <i>High_IC</i>	−0.0049** (−2.632)	−0.0051** (−2.491)
<i>High_IC</i>	0.0095 (1.049)	0.0075 (0.947)
<i>Treat</i> $\times$ <i>Post</i>	−0.0031 (−1.617)	−0.0035* (−1.940)
<i>Cash_Hold</i>	0.0018 (0.382)	0.0045 (0.796)
<i>Capex</i>	0.0191 (1.443)	0.0269 (1.459)
<i>Market-to-Book</i>	0.0007 (1.027)	0.0018** (2.267)
<i>Debt</i>	0.0056 (1.417)	0.0059 (1.284)
<i>Size</i>	0.0025 (0.585)	−0.0064 (−1.153)
$\log_Age$	0.0075*** (4.655)	0.0032** (2.064)
<i>Idiosyncratic Volatility</i>	−0.0179 (−0.859)	−0.0158 (−0.514)
<i>StateGDP_Growth</i>	−0.0258 (−1.385)	−0.0250 (−1.096)
$\log_StatePopulation$	−0.0047 (−0.133)	0.0203 (0.641)
Headquarter state FE	Yes	Yes
Industry-by-year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	29,157	29,157
Adjusted R^2	0.279	0.309

Notes. This table shows whether the effect of IC usage on operating leverage and payout policy is more pronounced for the industries with a relatively high proportion of independent contractors in a workforce. *High_IC* is an indicator equal to one if a firm in year t is in one of the five industries (“construction,” “professional and business services,” “other services,” “retail trade,” and “education and health services”) with relatively intensive IC usage and zero otherwise, according to the 2005 BLS Contingent and Alternative Employment Arrangements survey. Panel A presents how operating leverage changes in response to the 2004 MICL adoption, following Eisfeldt and Papanikolaou (2013). In panel A, the dependent variable, $\Delta \log(\text{EBIT})$, is the annual change in the natural log of earnings before interest and taxes for each firm in year t . Panel B shows the corresponding results on corporate payouts. We take share repurchases and total payout scaled by book assets as the dependent variables, respectively, in panel B. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

in stock repurchases and total payout after the MICL adoption is more pronounced for firms that are relatively close to bankruptcy (low Z score). In sum, Table 8

Table 8. Financial Constraints and Payout Policy

	(1) <i>Repurchases over Assets</i>	(2) <i>Total Payout over Assets</i>	(3) <i>Repurchases over Assets</i>	(4) <i>Total Payout over Assets</i>	(5) <i>Repurchases over Assets</i>	(6) <i>Total Payout over Assets</i>
<i>Treat × Post × High HP</i>	−0.0059*** (−3.000)	−0.0067*** (−3.125)				
<i>Treat × Post × High WW</i>			−0.0071*** (−6.357)	−0.0091*** (−8.712)		
<i>Treat × Post × Low Z</i>					−0.0026** (−2.457)	−0.0032** (−2.438)
<i>High HP</i>	0.0018 (0.644)	0.0021 (0.619)				
<i>High WW</i>			−0.0013 (−0.893)	−0.0025 (−1.401)		
<i>Low Z</i>					−0.0041*** (−3.043)	−0.0054*** (−3.389)
<i>Treat × Post</i>	−0.0008 (−0.308)	−0.0008 (−0.309)	0.0002 (0.097)	0.0009 (0.506)	−0.0032 (−1.593)	−0.0035* (−1.870)
<i>Cash_Hold</i>	0.0016 (0.345)	0.0043 (0.771)	0.0017 (0.360)	0.0045 (0.780)	0.0010 (0.219)	0.0036 (0.624)
<i>Capex</i>	0.0192 (1.461)	0.0270 (1.471)	0.0193 (1.454)	0.0270 (1.462)	0.0185 (1.404)	0.0260 (1.418)
<i>Market-to-Book</i>	0.0006 (1.057)	0.0017** (2.398)	0.0007 (1.109)	0.0019** (2.322)	0.0005 (0.873)	0.0016** (2.146)
<i>Debt</i>	0.0057 (1.374)	0.0061 (1.260)	0.0054 (1.361)	0.0058 (1.239)	0.0115** (2.508)	0.0136** (2.653)
<i>Size</i>	0.0032 (0.872)	−0.0055 (−1.100)	0.0012 (0.290)	−0.0087 (−1.537)	0.0010 (0.248)	−0.0082 (−1.458)
<i>log_Age</i>	0.0077*** (4.213)	0.0035* (1.964)	0.0072*** (4.363)	0.0027* (1.843)	0.0075*** (4.579)	0.0032* (2.004)
<i>Idiosyncratic Volatility</i>	−0.0199 (−0.933)	−0.0180 (−0.564)	−0.0189 (−0.902)	−0.0170 (−0.546)	−0.0188 (−0.892)	−0.0168 (−0.537)
<i>StateGDP_Growth</i>	−0.0262 (−1.411)	−0.0254 (−1.118)	−0.0260 (−1.407)	−0.0251 (−1.109)	−0.0265 (−1.436)	−0.0259 (−1.138)
<i>log_StatePopulation</i>	−0.0032 (−0.091)	0.0218 (0.686)	−0.0034 (−0.098)	0.0218 (0.696)	−0.0041 (−0.117)	0.0207 (0.662)
Headquarter state FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,157	29,157	29,157	29,157	29,157	29,157
Adjusted R ²	0.279	0.309	0.279	0.309	0.279	0.310

Notes. This table examines how firm-level financial constraints affect the relation between the curbed use of independent contractors and corporate payout policy. We use share repurchases and total payout scaled by book assets as the dependent variables. *High HP*, *High WW*, and *Low Z* are the indicators equal to one if a firm in year *t* has an above-median value for the Size and Age Index (Hadlock and Pierce 2010) and the Whited and Wu (2006) index and a below-median value for the Altman Z score, respectively, and zero otherwise. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute *t* statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

implies that financially constrained or distressed firms react sensitively to the law adoption by reducing their payout.

Third, we examine whether the decrease in corporate payouts is more pronounced for those with high operating leverage. Operating leverage is beneficial because it magnifies profits when sales increase, but it can be detrimental

because it also increases a firm’s probability of financial distress. Chen et al. (2019) argue that managers can anticipate such a trade-off. Thus, in normal times, a firm is likely to set its operating leverage at an optimal level because of the trade-off. However, the MICL acts as an exogenous shock that increases a firm’s operating leverage above this optimal level. If so, the resulting, higher-than-optimal

Table 9. Operating Leverage and Payout Policy

	(1) Repurchases Assets	(2) Total Payout/Assets
<i>Treat</i> × <i>Post</i> × <i>Op_Lev</i>	−0.0023* (−1.990)	−0.0039*** (−2.938)
<i>Op_Lev</i>	0.0085*** (5.208)	0.0108*** (5.606)
<i>Treat</i> × <i>Post</i>	−0.0020 (−0.769)	−0.0011 (−0.427)
<i>Cash_Hold</i>	0.0062 (1.307)	0.0101* (1.810)
<i>Capex</i>	0.0129 (0.912)	0.0189 (0.958)
<i>Market-to-Book</i>	−0.0000 (−0.021)	0.0009 (1.295)
<i>Debt</i>	0.0063 (1.538)	0.0069 (1.449)
<i>Size</i>	0.0125*** (2.843)	0.0064 (1.048)
<i>log_Age</i>	0.0074*** (4.546)	0.0031* (1.955)
<i>Idiosyncratic Volatility</i>	−0.0375* (−1.696)	−0.0407 (−1.300)
<i>StateGDP_Growth</i>	−0.0271 (−1.487)	−0.0266 (−1.199)
<i>log_StatePopulation</i>	−0.0069 (−0.190)	0.0171 (0.526)
Headquarter state FE	Yes	Yes
Industry-by-year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	29,157	29,157
Adjusted R ²	0.281	0.312

Notes. This table shows how firm-level operating leverage changes the effect of the Massachusetts Independent Contractor Law on payout policy. Because we need a firm-year operating leverage measure (*Op_Lev*), we follow Novy-Marx (2011), which defines operating leverage as the sum of cost of goods sold and selling, general, and administrative costs scaled by book assets. The dependent variables are share repurchases and total payout scaled by asset for a firm in year *t*. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute *t* statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; ***statistical significance at the 1% level.

operating leverage for treated firms may induce a manager to be more concerned about potential financial distress; such concern should be more pronounced for firms with high operating leverage. Thus, the decrease in payout because of the curbed IC usage is likely to be more pronounced for firms with already high operating leverage.

Because the earnings-sales sensitivity is a regression coefficient not available at the firm-year level, we instead follow Novy-Marx (2011), which defines a firm's

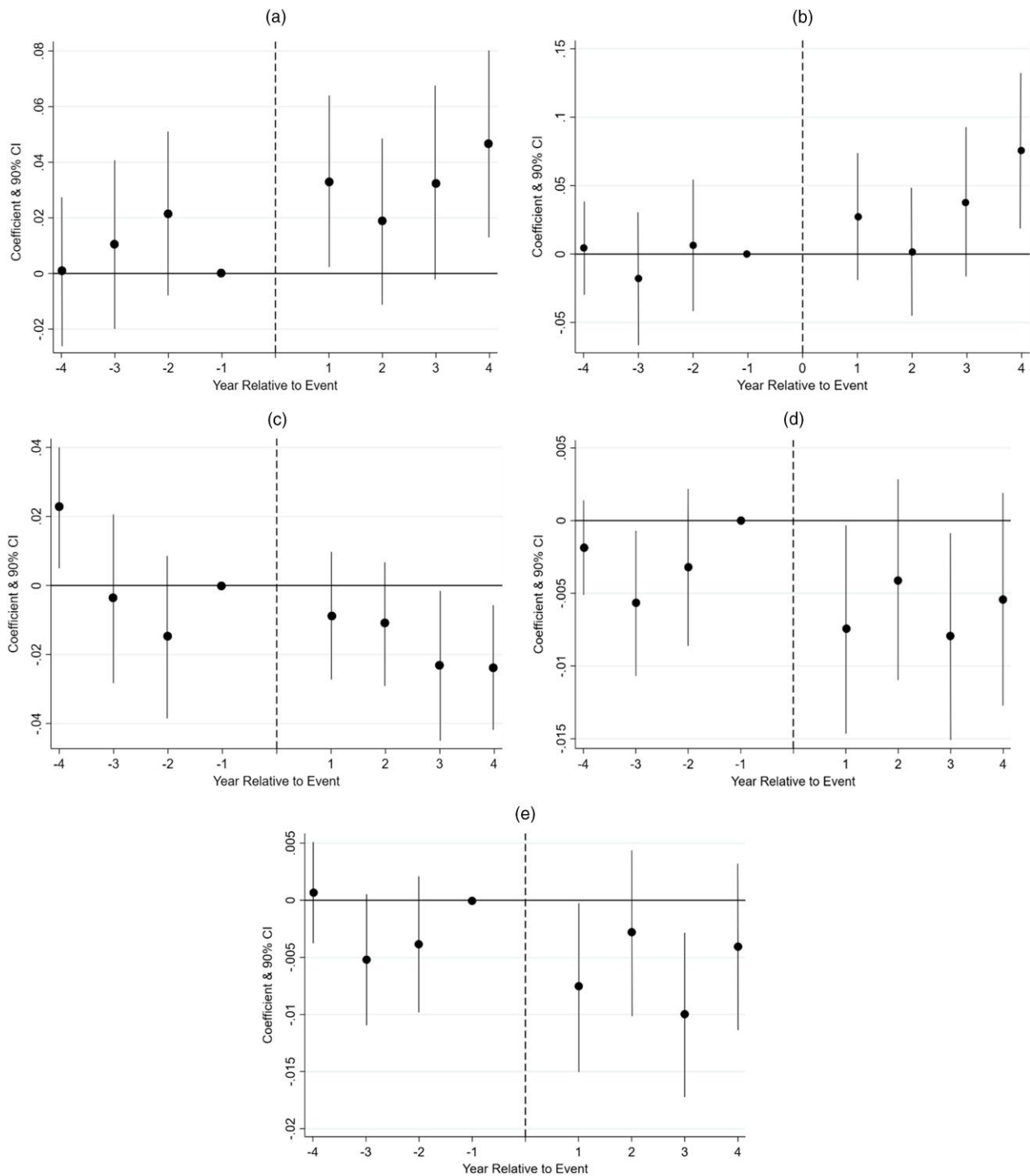
operating leverage in year *t* (*Op_Lev*) as annual operating costs (COGS plus SG&A) divided by book assets.²⁸ A higher *Op_Lev* implies higher operating leverage. In columns (1) and (2) in Table 9, we interact *Op_Lev* with *Treat* × *Post* to see if the treatment effect depends on firm-level operating leverage. Although the coefficient of *Treat* × *Post* is statistically indistinguishable from zero in both columns, the coefficient of *Treat* × *Post* × *Op_Lev* is negative and significant for both stock repurchases and total payout. In sum, Table 9 implies that firms with high operating leverage are more concerned about the stricter requirement for using ICs and reduce payout to a greater extent than firms with low operating leverage.

4.4. Robustness Tests

We next introduce a series of alternative specifications to show that our results remain robust. We first examine the timing of the treatment effect on firm-level outcomes to show that our findings are not driven by the extension of the pretreatment trend. To do so, we augment Equation (1) with lags and leads of the *Treat* × *Post* term. Specifically, we disaggregate the *Post* variable into indicators corresponding to different years, before and after the adoption of the MICL, and interact each indicator with *Treat*. Each interaction term (*Treat* × *Post*_{*t*}; −4 to +4) captures the difference in a dependent variable between the treated and the control group each year. Similar to Mirenda et al. (2022), we take the year *t* − 1, one year before the treatment, as the reference category such that the coefficient estimate for the year *t* − 1 is normalized to zero and other estimates are relative to the year *t* − 1.²⁹ The dependent variables are labor expenses (proxied by SG&A), operating costs, profitability, repurchases, and total payout. We regress each dependent variable on each *Treat* × *Post*_{*t*} term, the firm- and state-level controls, and the fixed effects (firm, industry by year, and headquarter state).

In panels (a) and (b) of Figure 1, none of the pretreatment coefficients significantly deviate from zero with labor expenses and operating costs as a dependent variable, respectively. Thus, there seems to be no divergent pattern between the treated group and the control group before the treatment in terms of costs. In the posttreatment period, the coefficient estimate increases relative to the year *t* − 1 estimate. The coefficient of *Treat* × *Post*_{*t*} is positive and significant in the first and fourth years of the posttreatment period for labor expenses and in the fourth year for operating costs.

In panel (c), the pretreatment trend is constant for the three years before the MICL adoption when profitability is used as a dependent variable. In the posttreatment period, the coefficient estimate decreases relative to the year *t* − 1 estimate. The coefficient of *Treat* × *Post*_{*t*} is negative and significant in the third and fourth years of the posttreatment period for operating profitability.

Figure 1. (Color online) Timing of the Treatment Effect

Notes. This figure shows the timing of the treatment effect. We disaggregate the *Post* indicator into each dummy variable ($Post_t$, $t = -4$ to 4) and interact it with *Treat*. Each interaction term, $Treat \times Post_t$, captures the difference in a dependent variable each year between the treated group and the control group. Similar to Miranda et al. (2022), we take the year $t = -1$, one year before the treatment, as the reference category. So, the coefficient estimate for $t = -1$ is normalized to zero, and other estimates are relative to the year $t = -1$. The dependent variables are SG&A, operating costs (COGS plus SG&A), operating income before depreciation and amortization, stock repurchases, and total payout scaled by book assets. We regress each dependent variable on each $Treat \times Post_t$ term, the firm- and state-level controls, and fixed effects (firm, industry by year, and headquarter state). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. A vertical line for each coefficient shows the 90% confidence interval (90% CI). Appendix A presents the variable definitions. (a) Labor expenses (proxied by SG&A). (b) Operating costs. (c) Profitability. (d) Stock repurchases. (e) Total payout.

Table 10. Entropy Balancing

Panel A: Before reweighting				
	Treated		Control	
	Mean	Variance	Mean	Variance
Cash_Hold	0.3726	0.0801	0.2221	0.0604
Capex	0.0437	0.0024	0.0533	0.0037
Market-to-Book	2.4900	4.1530	2.1570	3.2930
Debt	0.1363	0.0418	0.2058	0.0476
Size	0.2870	0.0690	0.3303	0.0852
log_Age	2.1960	0.8513	2.3310	0.9418
Idiosyncratic Volatility	0.0439	0.0006	0.0396	0.0006
Panel B: After reweighting				
	Treated		Control	
	Mean	Variance	Mean	Variance
Cash_Hold	0.3726	0.0801	0.3723	0.0801
Capex	0.0437	0.0024	0.0437	0.0024
Market-to-Book	2.4900	4.1530	2.4890	4.1510
Debt	0.1363	0.0418	0.1364	0.0418
Size	0.2870	0.0690	0.2871	0.0690
log_Age	2.1960	0.8513	2.1960	0.8516
Idiosyncratic Volatility	0.0439	0.0006	0.0439	0.0006
Panel C: Results				
	(1) Labor Expenses/Assets	(2) Profitability	(3) Repurchases/Assets	(4) Total Payout/Assets
Treat × Post	0.0202*** (2.916)	−0.0133** (−2.400)	−0.0032* (−1.723)	−0.0051*** (−3.886)
Cash_Hold	−0.2032*** (−9.585)	0.0498* (1.892)	−0.0047 (−1.038)	−0.0041 (−0.681)
Capex	0.4582*** (4.747)	−0.2401* (−1.723)	0.0161** (2.248)	0.0167 (1.422)
Market-to-Book	0.0351*** (14.937)	−0.0214*** (−19.825)	0.0004 (0.905)	0.0010*** (2.682)
Debt	0.0024 (0.083)	−0.1118*** (−6.630)	−0.0038* (−1.917)	−0.0045** (−2.526)
Size	−0.6589*** (−23.230)	0.5447*** (12.838)	−0.0011 (−0.282)	−0.0100* (−1.943)
log_Age	0.0106* (1.868)	0.0409*** (5.768)	0.0078*** (6.598)	0.0016 (0.842)
Idiosyncratic Volatility	0.5860 (1.504)	−1.0074*** (−4.291)	−0.0291* (−1.779)	−0.0255 (−1.044)
StateGDP_Growth	0.1992** (2.646)	−0.0473 (−0.612)	−0.0231 (−0.859)	−0.0060 (−0.158)
log_StatePopulation	0.1195 (0.790)	0.0361 (0.249)	0.0008 (0.019)	0.0247 (0.658)
Headquarter state FE	Yes	Yes	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	29,157	29,157	29,157	29,157
Adjusted R ²	0.767	0.801	0.297	0.321

Notes. This table shows the results based on entropy balancing (Hainmueller 2012). We reweight the observations in the control group such that the firm-level covariates in the regressions are balanced on the mean and the variance. Panels A and B compare the covariate balance across the treated group and the control group before and after reweighting. Panel C presents the results after taking entropy balancing into account. We use SG&A, operating income, share repurchases, and total payout scaled by book assets as the dependent variables, respectively. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute *t* statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

Table 11. Staggered Difference-in-Differences

	(1) <i>Labor Expenses/Assets</i>	(2) <i>Profitability</i>	(3) <i>Repurchases/Assets</i>	(4) <i>Total Payout/Assets</i>
<i>Treat</i> × <i>Post</i>	0.0200*** (3.837)	−0.0142*** (−3.289)	−0.0044** (−2.574)	−0.0051*** (−3.023)
<i>Cash_Hold</i>	−0.1475*** (−8.519)	0.0197 (1.349)	0.0017 (0.361)	0.0044 (0.779)
<i>Capex</i>	0.4042*** (4.068)	−0.2838** (−2.491)	0.0191 (1.450)	0.0269 (1.460)
<i>Market-to-Book</i>	0.0294*** (16.066)	−0.0133*** (−6.089)	0.0007 (1.046)	0.0018** (2.283)
<i>Debt</i>	−0.0401** (−2.338)	−0.0992*** (−7.392)	0.0054 (1.373)	0.0058 (1.258)
<i>Size</i>	−0.5173*** (−9.972)	0.4399*** (10.229)	0.0023 (0.562)	−0.0065 (−1.182)
<i>log_Age</i>	0.0101* (1.757)	0.0180*** (2.908)	0.0075*** (4.609)	0.0032** (2.037)
<i>Idiosyncratic Volatility</i>	0.9047*** (5.587)	−1.0957*** (−4.655)	−0.0185 (−0.887)	−0.0165 (−0.536)
<i>GDP_Growth</i>	0.0699 (1.082)	−0.0031 (−0.048)	−0.0257 (−1.390)	−0.0250 (−1.102)
<i>log_Population</i>	0.1401 (1.323)	−0.0357 (−0.472)	−0.0041 (−0.118)	0.0202 (0.644)
Headquarter state FE	Yes	Yes	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	29,157	29,157	29,157	29,157
Adjusted R^2	0.801	0.794	0.279	0.309

Notes. This table shows that our results are robust to using a staggered DID design using the independent contractor misclassification statutes that went into effect in Massachusetts, New Mexico, Oregon, and New Jersey from 2004 to 2007. The original test is based on the 2004 Massachusetts Independent Contractor Law adoption. The sample period spans from 2000 to 2007. We take labor-related expense (SG&A), operating income before depreciation and amortization, share repurchases, and total payout scaled by book assets as the dependent variables, respectively. We include firm, industry-by-year, and headquarter state fixed effects (FEs). We compute t statistics (in parentheses) using robust standard errors clustered at the state level. The constant is not shown for brevity. Appendix A presents the variable definitions.

*Statistical significance at the 10% level; **statistical significance at the 5% level; ***statistical significance at the 1% level.

In panels (d) and (e), when stock repurchases and total payout are used as a dependent variable, respectively, the pretreatment trend is largely constant. Although the coefficient estimate for stock repurchases in the year $t - 3$ is significant, total payout does not show any significant estimate in the pretreatment period. In the posttreatment period, the coefficient estimate decreases relative to the year $t - 1$ estimate. The coefficient of $Treat \times Post_t$ is negative and significant in the first and third years of the posttreatment period for both stock repurchases and total payout. Overall, Figure 1 shows that the changes in labor expenses, operating costs, profitability, repurchases, and payout after the MICL adoption do not seem to be driven by the extension of the pretreatment trend.

We also address the concern that our inferences may be driven by differences in firm fundamentals between the treated group and the control group. To do so, we follow an entropy-balancing approach (Hainmueller 2012, Hainmueller and Xu 2013). This method calculates and assigns weights to each observation in the control group

so that the moments (e.g., mean and variance) of covariates match between the treated group and the control group. Regarding covariates, we match on the firm-level controls used in the baseline regression. We require that the entropy balancing approach compute weights for the observations in the control group in a way that the sample mean and variance of each firm-level control match those in the treated group. Panels A and B of Table 10 present the mean and variance of the firm-level controls for the two groups. After reweighting, matching in the sample mean and variance is almost exact. With the reweighted sample, we re-estimate Equation (1) and see the variation in labor expenses (proxied by SG&A), profitability, and corporate payouts for the treated firms relative to the control after the MICL adoption. Panel C of Table 10 corroborates our results, showing the increase in SG&A and the decrease in profitability, stock repurchases, and total payout for the treated firms following the 2004 MICL. In sum, the treated group's exposure to curbed IC usage still reduces its profitability and corporate payouts even

after we make the difference in observable fundamentals between the two groups insignificant.³⁰

We also adopt a staggered DID design that includes similar laws adopted in other states during our sample period. In our baseline specification, we focus on the Massachusetts regulatory shift because it is arguably the most binding regulation among the similar U.S. state-level statutes for several reasons (Keohane and Schap 2021). First, the ABC test adopted by the state of Massachusetts is stricter than the two other types of independent contractor misclassification regulation (the common law test and the IRS test). There are three prongs that reflect control, usual course of business, and independence in the ABC test. The common law test only concerns the “control” part of the ABC test, and the IRS test is less binding than the ABC test. Second, the ABC test in Massachusetts is stricter than the comparable ABC test in other U.S. states because of the modified “B” prong for Massachusetts (explained in Section 2.3).³¹ Because the treatment effect of IC misclassification statutes on corporate policies is likely to be most pronounced when the regulation is the most binding, we focus on the MICL, which offers the strictest requirements to classify a worker as an IC.

Nevertheless, following Deknatel and Hoff-Downing (2015), we use the state-level statutes adopted in Massachusetts, New Mexico (effective from June 2005), Oregon (effective from January 2006), and New Jersey (effective from July 2007) and check whether our findings remain robust to the staggered DID design.³² In Table 11, we follow Equation (1), except that we assign treatment slightly differently. *Treat* is now equal to one if a firm is headquartered in Massachusetts, New Mexico, Oregon, or New Jersey and zero otherwise. New Mexico and New Jersey firms also have to be a construction company (NAICS 23) to be treated because the two states only targeted construction firms. Thus, $Treat \times Post$ is equal to one if a firm’s fiscal year end comes after the effective date of the IC misclassification statute for each state in which a treated firm is headquartered and zero otherwise. In column (1), we document that labor expenses (proxied by SG&A) increase significantly, which reduces a firm’s profitability (column (2)). Columns (3) and (4) corroborate that the treated firms reduce their corporate payouts. In sum, taking into account similar law adoptions in other states in the sample period does not nullify our results.³³

We also extend the sample period to 2017 and use the staggered DID design.³⁴ We follow the “definitional” statute adoptions in 15 U.S. states from 2004 to 2012, documented by Deknatel and Hoff-Downing (2015).³⁵ If a state changes the “enforcement” statute alone (e.g., increasing monetary penalties), then firms complying with the current definitional requirement for IC classification should not be affected. By imposing a stricter IC definition on employers, the statute makes it harder for

firms using ICs to maintain their classification, regardless of whether a firm has complied with the previous IC regulation or not.

In Table OA6 in the online appendix, *Treat* is an indicator variable equal to one if a firm in year t is in 1 of 15 states that changed its definitional statutes from 2004 to 2012 and zero otherwise. If a statute is industry specific, only firms in the affected industries are treated. *Post* is equal to one if a firm-year observation for a treated firm is in a year after the adoption of the IC misclassification statute for each state and zero otherwise. We find a firm’s profitability, stock repurchases, and total payout decrease after the adoption of the state-level IC misclassification statutes. We find that labor expenses increase, albeit not significantly. The relatively weak IC misclassification statutes adopted in this later period may subdue statistical significance.³⁶ Thus, we focus on the period surrounding the MICL adoption to examine the treatment effect of curbed IC usage on operating performance and payout because the Massachusetts statute is likely more binding than the employment classification tests enacted in other states.

Goodman-Bacon (2021) argues that in the standard staggered DID design, the DID estimate may be biased when the magnitude of the treatment effect varies across units (e.g., firms or states) or over time. In our context, the stringency of the IC classification statute likely varies across U.S. states. Even within a state, relatively minor regulatory adjustments after the law adoption can make the treatment effect vary over time. To address this concern, Gardner (2022) proposes a two-stage methodology. In the first stage, one regresses an outcome variable on group and period fixed effects using only the untreated observations. In the second stage, one subtracts the estimated group and period fixed effects from an observed outcome, and the adjusted outcome variable is then regressed on the treatment variable of interest. Gardner (2022) argues that this methodology isolates the average treatment effect on the treated, even with heterogeneous treatment effects across units or over time. Using the methodology of Gardner (2022), we test whether the results in Table 11 and Table OA6 in the online appendix are robust to addressing the bias from heterogeneous treatment effects.³⁷

In panel A of Table OA7 in the online appendix, we focus on the law adoptions in Massachusetts, New Mexico, Oregon, and New Jersey during the period from 2000 to 2007. In panel B, we use the sample period from 2000 to 2017 using the changes in the definitional statutes used in Table OA6 in the online appendix. The dependent variables are labor-related expenses (proxied by SG&A), profitability, share repurchases, and total payout. In the first stage, we include the firm- and state-level controls and the fixed effects. In the second stage, we only consider the $Treat \times Post$ term. In panel A, we find that the increase in labor expenses (proxied by SG&A)

and the decrease in profitability, stock repurchases, and total payout are statistically significant; in panel B, only the increase in labor expenses and the decrease in profitability are significant. We conclude that although the bias because of heterogeneous treatment effects subdues some of the findings in the extended period when weaker IC statutes were enacted, the results using the period from 2000 to 2007 are robust to addressing the bias.

Lastly, we note the results of several additional robustness checks. In Table OA8 in the online appendix, we scale stock repurchases by market value of assets, sales, and cash flow and find a significant decline in stock repurchases after the MICL adoption. We also try the payout flexibility measure of Bonaimé et al. (2014); the coefficient of $Treat \times Post$ is negative, albeit statistically insignificant. In Table OA9 in the online appendix, we exclude observations in 2006 and 2007 to check whether this increase in the minimum wage makes a firm's labor cost structure more fixed and thus, may drive our findings.³⁸ Our findings remain robust. In Table OA10 in the online appendix, we omit construction firms from the full sample to address the concern that Massachusetts construction firms are more likely to have been aware of the regulatory change than firms in other industries. The results remain unchanged. In Tables OA11 and OA12 in the online appendix, we exclude the time-varying firm- and state-level controls to show that the choice of control variables does not significantly affect our findings.

5. Conclusion

We exploit the 2004 Massachusetts Independent Contractor Law to investigate the effect of ICs on a firm's operations and payout policy. The law exogenously decreases Massachusetts firms' IC usage by imposing stricter classification requirements for ICs. Because ICs offer labor flexibility and cost savings compared with employees, Massachusetts firms experience increases in labor-related expenses and decreases in profits after the law adoption. Treated firms' earnings become more sensitive to changes in sales after the exogenous reduction in IC usage, signaling higher labor-induced operating leverage. As a result, treated firms become more conservative with their payout policy. We find evidence that

the adoption of the 2004 MICL reduces a firm's total payout, especially stock repurchases. The economic magnitude is large; firms decrease stock repurchases by 22% relative to the sample mean.

Our additional tests highlight the trade-off between operating risk and financial risk. We find that the increase in operating leverage and the decrease in corporate payouts are more pronounced for (1) firms with intensive IC usage, which are most impacted by the law; (2) financially constrained or distressed firms, which need more internal funds to withstand adverse shocks; and (3) firms with higher operating leverage.

Although previous studies have focused on how regular employment affects corporate financial policies, the literature offers little evidence on how nonregular employment influences firm outcomes, albeit with a few exceptions (e.g., Chino 2021, Hahn et al. 2021, Kuzmina 2023). Given the increase in the proportion of ICs in the U.S. workforce during the twenty-first century, studying how nonregular employment affects a firm's other financial or nonfinancial aspects can be fruitful for future research. Considering the increased attention on share repurchases in recent years and their impact on wages and investment, politicians and regulators should also be interested in how a firm's use of independent contractors affects payout policy.

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Appendix A. Variable Definitions

Table A.1. Variable Definitions

Variable	Definition (Compustat variable name in parentheses)
Dependent variables	
<i>Dividends over Assets</i>	Cash dividend (DV) over total assets (AT)
<i>Repurchase over Assets</i>	Purchase of common and preferred stock (PRSTKC) minus the reduction in the book value of preferred stock (PSTKRV), scaled by total assets (AT)
<i>Total Payout over Assets</i>	Sum of repurchases (PRSTKC minus the reduction in PSTKRV) and cash dividends (DV), scaled by total assets (AT)
<i>Repurchases over MVA</i>	Purchase of common and preferred stock (PRSTKC) minus the reduction in the book value of preferred stock (PSTKRV), scaled by market value of assets (AT minus CEQ plus PRCC_F $\times$ CSHO)
<i>Repurchases over Sale</i>	Purchase of common and preferred stock (PRSTKC) minus the reduction in the book value of preferred stock (PSTKRV), scaled by sales (SALE)
<i>Repurchases over CashFlow</i>	Purchase of common and preferred stock (PRSTKC) minus the reduction in the book value of preferred stock (PSTKRV), scaled by operating income before depreciation and amortization (OIBDP)
<i>Payout Flexibility</i>	Stock repurchases (PRSTKC minus the reduction in PSTKRV) scaled by total payout (the sum of DV and stock repurchases), following Bonaimé et al. (2014)
$\Delta \log(\text{EBIT})$	The annual change in the natural log of earnings before interest and taxes (EBIT) for each firm in year t
<i>Labor Expenses over Assets</i>	Because of a lack of data on staff expenses (XLR), we use selling, general, and administrative expense (XSGA) scaled by book asset (AT) as a proxy for labor expense (e.g., Lee et al. 2018); we set SG&A to zero if missing
<i>Operating Costs</i>	The sum of cost of goods sold (COGS) and SG&A (XSGA) scaled by book assets (AT)
<i>Legal Settlements</i>	Using AuditAnalytics Litigation database, we aggregate the sum of legal settlements for each firm in a year and scale it by revenue (SALE)
<i>Profitability</i>	Operating income before depreciation and amortization (OIBDP) scaled by book asset (AT)
$\log(1 + \text{Employees})$	The natural log of 1 plus the number of employees (EMP) for a firm in year t
Main independent variables	
<i>Treat</i>	An indicator variable equal to 1 if a firm is headquartered in Massachusetts and 0 otherwise
<i>Post</i>	An indicator variable equal to 1 if a firm-year's fiscal year-end comes on June 2004 or onward and 0 otherwise
Control variables	
<i>Cashflow</i>	Operating income before depreciation (OIBDP) over book asset (AT)
<i>Size</i>	Fraction of NYSE firms with a lower or equal market capitalization in the same year; market capitalization used in computing NYSE percentile is stock price times the number of outstanding shares (PRC times SHROUT)
<i>Capex</i>	Capital expenditure (CAPX) over total assets (AT)
<i>Idiosyncratic volatility</i>	Standard deviation of the residuals from a regression of the daily stock return in excess of the risk-free rate on the market factor based on the value-weighted market return (source: CRSP and Kenneth French's website); daily returns over the fiscal year are used
$\log_Age$	Log of 1 plus the number of years since the firm's first appearance in CRSP
<i>Market-to-book</i>	The market value of assets (book assets (AT) plus market value of equity (PRCC_F $\times$ CSHO) minus book value of equity (CEQ)) scaled by book assets (AT)
<i>Debt</i>	Long-term debt (DLTT) plus debt in current liabilities (DLC), scaled by total assets (AT)
<i>Cash_hold</i>	Cash and short-term investments (CHE) over total assets (AT)
<i>StateGDP_Growth</i>	Annual growth rate in real GDP for a state in year t
$\log_StatePopulation$	The natural log of state population (in thousands) in a year
Conditioning variables	
<i>Op_Lev</i>	Firm-year measure of operating leverage following Novy-Marx (2011), which is defined as the sum of cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA) scaled by book assets
<i>Change in log (Sale)</i>	Annual change in the log of sales (SALE)
<i>High WW</i>	Indicator equals 1 when the Whited and Wu index is greater than the median value for the full sample and otherwise 0 (Whited and Wu 2006). We define the WW index as follows: $-0.091 \text{ times } ((IB + DP)/AT) \text{ minus } 0.062 \text{ times indicator for positive dividends plus } 0.021 \text{ times } DLTT/AT \text{ minus } 0.044 \text{ times the natural logarithm of assets plus } 0.102 \text{ times industry sales growth minus } 0.035 \text{ times firm sales growth}$. The indicator for positive dividends equals 1 when common dividends (DVC) plus preferred dividends (DVP) is strictly greater than 0. Firm sales growth is the relative change in SALE. Industry sales growth is the mean of the firm sales growth for the three-digit SIC industry to which the firm belongs.
<i>High HP</i>	Indicator equals 1 when the Size and Age index is greater than the median value for the full sample and otherwise 0 (Hadlock and Pierce 2010). We compute the HP index as follows: $-0.737 \text{ times the natural logarithm of inflation-adjusted (at 2004 price level) book assets (AT)}$

Table A.1. (Continued)

Variable	Definition (Compustat variable name in parentheses)
<i>Low_Z</i>	plus 0.043 times the squared value of the log of inflation-adjusted book assets minus 0.04 times firm age. Firm age is the number of years since the firm's first appearance in Compustat. The inflation-adjusted book asset is capped at 4,500, and the firm age is capped at 37. Indicator equals 1 when the Altman Z score is lower than the median value for the full sample and otherwise 0; we compute the Z score as 3.3 times (OIADP/AT) plus 1.2 times (ACT – LCT)/AT plus (SALE/AT) plus 0.6 times (PRCC_F × CSHO)/(DLTT + DLC) plus 1.4 times (RE/AT) for a firm each year
<i>High_IC</i>	An indicator equal to 1 if a firm in year <i>t</i> is in construction (NAICS 23), professional and business activities (NAICS 54–56), other services (NAICS 81), retail trade (NAICS 44 and 45), or education and health services (NAICS 61 and 62) industry and 0 otherwise
<i>US_MNC</i>	An indicator equal to 1 if a firm in year <i>t</i> reports positive foreign taxes and 0 otherwise

Appendix B. Example of IC Misclassification-Related Lawsuits: Amero v. Townsend Oil Co.

Townsend Oil Co. is a firm headquartered in Danvers, Massachusetts, and it distributes oil and related products around northeastern Massachusetts.³⁹ Its busy season comes in the winter, and the firm recruits additional drivers. Plaintiff Hughes Amero started driving an oil delivery truck as an independent contractor in December 2000. Although the company's employee drivers received hourly and overtime pay, its ICs were not granted overtime pay, received \$0.09 per gallon of oil delivered, and had to pay for their own vehicle and insurance. Later in 2005, Amero got injured while refueling his vehicle. He requested that the firm compensate him for medical expenses and for the cost of finding a replacement driver, but Townsend Oil Co. denied the request. Later in the same year, the two terminated their contract.

In June 2007, Amero sued the company in Superior Court. The plaintiff argued that Townsend Oil Co. violated the MICL by wrongly classifying him as an independent contractor because his service was related to the core business of Townsend Oil Co. and he was under the company's direction and control. The court confirmed that Amero had been misclassified as an IC. According to the judge's decision, Amero's service was significantly related to the core business of Townsend Oil Co. (oil delivery). The firm also required Amero to wear a company uniform, paint Townsend's logo on Amero's truck, and sign a noncompetition agreement, which put Amero under the company's control. Thus, Amero did not meet the requirement for IC classification based on the 2004 MICL and thus, should have been classified as an employee. As a result, Townsend Oil Co. incurred monetary liability related to Amero's foregone overtime pay.

Endnotes

¹ According to the May 2005 survey by the Bureau of Labor Statistics (BLS), independent contractors (10.3 million people in the United States) are larger in number than the other nonregular employment groups, such as on-call workers (2.5 million people), temporary help agency workers (1.2 million people), and workers from contract firms (813,000 people).

² ICs may not be better than employees in every aspect. For example, employees may be more loyal and easier to control from management's perspective. However, if firms optimize and strike a

balance between ICs and employees based on their business model, then suddenly imposing exogenous constraints on the use of ICs may harm firms.

³ Similar to Lee et al. (2018) and Gutiérrez et al. (2020), we use SG&A to proxy for labor-related expenses.

⁴ Operating disruption here means a deviation from normal business activities because of the unexpected exogenous constraints on the use of ICs. Blanchard and Landier (2002) argue that firms determine the use of nonregular workers based on the trade-off between labor-adjustment flexibility offered by nonregular workers and higher productivity offered by regular workers. Without any exogenous shock, a firm sets the share of ICs at an optimal level that maximizes profitability. The deviation from the optimal level because of the shock implies lower profitability than the optimal level.

⁵ Payout policy, particularly stock repurchases, has received much attention from politicians in recent years because of the concern that firms funnel most of their excess profits into buybacks, which could instead be used to increase wages and benefits or to make investments.

⁶ Appendix B presents an example of an IC misclassification lawsuit in Massachusetts: Amero v. Townsend Oil Co.

⁷ According to the IRS 20-factor test, workers are generally employees if they (1) must comply with the employer's instructions about the work; (2) receive training from or at the direction of the employer; (3) provide services that are integrated into the business; (4) provide services that must be rendered personally; (5) hire, supervise, and pay assistants for the employer; (6) have a continuing working relationship with the employer; (7) must follow set hours of work; (8) work full time for an employer; (9) do their work on the employer's premises; (10) must do their work in a sequence set by the employer; (11) must submit regular reports to the employer; (12) receive payments of regular amounts at set intervals; (13) receive payments for business and/or traveling expenses; (14) rely on the employer to furnish tools and materials; (15) lack a major investment in facilities used to perform the service; (16) cannot make a profit or suffer a loss from their services; (17) work for one employer at a time; (18) do not offer their services to the general public; (19) can be fired at any time by the employer; and (20) may quit work at any time without incurring liability. No individual factor is decisive, and the importance of each depends on the occupation of a worker and the context in which services are performed.

⁸ For example, if a misclassified IC has worked over 40 hours per week without receiving overtime pay (normally 1.5 times the regular pay rate), he or she may claim three times the amount owed over the three years before the misclassification was detected.

⁹ See <https://archives.lib.state.ma.us/bitstream/handle/2452/46769/ocn228503475.pdf> (accessed March 23, 2023).

¹⁰ For instance, the online article by Gilman Law LLP (see <https://gilmanlawllp.com/employment-law/massachusetts-independent-contractor-misclassification-lawsuit/>) (accessed March 23, 2023) states that “Massachusetts has some of the strictest laws in the country governing the classification of workers as independent contractors.” Another online article by Ogletree Deakins, the labor law firm (see <https://ogletree.com/insights/massachusetts-high-court-rules-prong-two-of-independent-contractor-test-is-severable-for-faaaa-preemption-purposes/>) (accessed March 23, 2023), states that “the independent contractor test used in applying Massachusetts wage statutes is arguably the most difficult such test in the nation.”

¹¹ We assume that the 2004 law adoption reduces the use of independent contractors for the affected firms because this cannot be directly measured because of data limitations (Bernhardt 2014). Law practitioners also share a concern that the MICL will deter firms from using independent contractors. See, for example, <https://www.morse.law/news/the-massachusetts-independent-contractor-law/> (accessed March 23, 2023).

¹² The data are available at <https://sraf.nd.edu/data/augmented-10-x-header-data> (accessed July 9, 2019).

¹³ We acknowledge that some firms in the control group may be headquartered outside Massachusetts, but their establishments may be in Massachusetts. Thus, the 2004 MICL may affect some control firms. However, those firms should bias the coefficient estimate toward zero because they should reduce the difference in outcomes between the treated group and the control group.

¹⁴ Although Oregon, New Mexico, and New Jersey also adopted a statute to address IC misclassification, we believe that the MICL is more binding than the three other statutes and thus, focus on the Massachusetts law. Later, we introduce a staggered DID design based on the four statutes and show that the results are robust.

¹⁵ Of the 29,157 firm-year observations in our sample, only 1,676 have nonmissing values for Compustat XLR. Also, for the observations that have nonmissing Compustat XSGA and XLR, the two variables show a highly positive and significant correlation ($\rho = 0.6087$).

¹⁶ The statute that amends the definition of IC (MASS. GEN. LAWS ANN. Ch. 149, 148B) was signed into law and became effective on July 19, 2004, and the bill that revises the enforcement statute (MASS. GEN. LAWS ANN. Ch. 149, 27C) was enacted into law on June 10, 2004. Thus, we start the posttreatment period from the end of June 2004. The result is robust to creating *Post* based on year 2004, the effective date of the MICL (July 2004), or omitting the year 2004 from the analysis to account for the possibility that the treated firms may need time to adjust their policy.

¹⁷ Firm fixed effects do not necessarily absorb headquarter state fixed effects because a firm may move its headquarter during the sample period. In Table OA1 in the online appendix, we show that excluding firms relocating their headquarters across states does not nullify our findings. Defining industry-by-year fixed effects using three-digit SIC codes does not nullify our results either (untabulated).

¹⁸ The correlation between the NYSE market-cap percentile and the log of book asset is highly positive and significant ($\rho = 0.8539$).

¹⁹ The BLS only presents cross-sectional data on nonregular employment at the broad industry level (e.g., two-digit NAICS) available for February of 1995, 1997, 1999, 2001, 2005, and May of 2017. To see the BLS sector aggregation titles using NAICS, see https://www.bls.gov/bls/naics_aggregation.htm (accessed March 23, 2023).

²⁰ In Table 2, the average operating costs scaled by book assets exceed one. It is not uncommon for a firm’s operating costs (COGS plus SG&A) to exceed the amount of book assets. In the sample of Novy-Marx (2011) from 1963 to 2008, at least 40% of the observations show that the average operating costs scaled by book assets are greater than one.

²¹ In Table OA2 in the online appendix, we test how the number of regular employees changes after the MICL adoption. After controlling for the observable firm- and the state-level characteristics, firm, year, and headquarters fixed effects, we do find a significant increase in the number of employees in industries that are more likely to rely on ICs. However, after additionally controlling for unobservable time-varying industry-level characteristics, the coefficient of *Treat* $\times$ *Post* $\times$ *High_IC* is positive but not significant. Hypothetically, firms can react to the MICL by (1) reclassifying ICs into employees or (2) stopping the use of the former ICs and contracting the workforce. The balance between (1) and (2) appears to yield mixed results when we test how the number of employees changes in response to the MICL adoption.

²² The 2005 New England In-House article states, “Employees also may file civil actions for themselves and others similarly situated seeking treble damages, attorneys’ fees and costs. This is a fertile area for claims, and recoverable money damages can be substantial.” See <https://www.morse.law/news/the-massachusetts-independent-contractor-law/> (accessed March 23, 2023).

²³ In Table OA3 in the online appendix, among the firms covered by the AuditAnalytics Litigation database, we find that the amount of legal settlements increases for Massachusetts firms relative to non-Massachusetts firms after the MICL adoption.

²⁴ See <https://www.morse.law/news/the-massachusetts-independent-contractor-law/> (accessed March 23, 2023).

²⁵ Chen et al. (2019) note that a finance textbook (e.g., Brealey et al. 2010) defines operating leverage as the sensitivity of profits to sales.

²⁶ In Table OA4 in the online appendix, we show that our results are not driven by the U.S. multinationals headquartered outside of Massachusetts that increase payout because of the American Jobs Creation Act (e.g., Faulkender and Petersen 2012).

²⁷ In untabulated analysis, we use a Tobit model and still find the significant decrease in stock repurchases after the law adoption.

²⁸ One may try regressing changes in earnings on changes in sales with a rolling window of observations and claim the resulting coefficient to be “a firm’s operating leverage in year t .” However, such measure is still relying on the observations prior to year t .

²⁹ Mirenda et al. (2022) do not present the confidence interval for the year $t - 1$ estimate.

³⁰ When we further match on mean, variance, and skewness, we still find the significant increase in labor expenses (proxied by SG&A) and the significant decrease in operating profitability and total payout for the treated group relative to the control. When examining the change in stock repurchases, we find the coefficient of *Treat* $\times$ *Post* to be negative but with a p -value of 0.11 (untabulated).

³¹ “Massachusetts law requires that the work done by the contractor be outside the core business of the employer. This is the part of the test that has proven to be the most problematic for businesses who use independent contractors.” See <https://www.slnlaw.com/independent-contractor-test.html> (accessed March 23, 2023).

³² Table OA5 in the online appendix shows that our results are robust to excluding observations in New Mexico, Oregon, and New Jersey.

³³ In untabulated analysis, we narrow the sample to the New England region (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont) and rerun the analysis. We find that profitability and stock repurchases significantly decrease after the 2004 MICL adoption. SG&A also shows a positive estimate, but it is statistically insignificant.

³⁴ We end the extended sample period in 2017 because the review by Keohane and Schap (2021) concluded in July 2017.

³⁵ “Definitional” statutes refer to state-level legislation that “altered their statutory definitions of independent contracting or employment

relationships,” and “enforcement” statutes refers to state-level legislation that strengthened or transformed “enforcement against the misclassification of workers” (Deknatel and Hoff-Downing 2015).

³⁶ Aside from the claim that the 2004 Massachusetts MCL is more binding than IC misclassification statutes adopted in other states, we argue that, starting in 2008, some U.S. states adopt “definitional” statutes that are weaker than the MCL. According to Deknatel and Hoff-Downing (2015), from 2004 to 2007 (during our sample period), Massachusetts, New Jersey, New Mexico, and Oregon strengthened their definitional requirements for IC classification. Massachusetts, New Jersey, and New Mexico adopted all three prongs of the ABC test, with the particularly strong B prong in Massachusetts. Oregon adopted the A and C prongs of the ABC test. From 2008 to 2012 (after our sample period), Delaware, Illinois, Kansas, Maine, Maryland, Minnesota, Nebraska, New Hampshire, New York, Pennsylvania, and Utah revised their definitional statutes. Although 6 of these 11 states (Delaware, Illinois, Maine, Maryland, Nebraska, and New Hampshire) adopted all three prongs of the ABC test, Minnesota and New York followed the common law test, which is primarily focused on the A prong of the ABC test (Keohane and Schap 2021). Kansas adopted the A and B prong, whereas Pennsylvania and Utah followed the A and C prongs. In summary, the proportion of U.S. states not adopting all three prongs of the ABC test is higher after 2007 (5 of 11) than from 2004 to 2007 (1 of 4), and the Massachusetts statute is particularly strict compared with the statutes adopted by other states.

³⁷ We use the “did2s” STATA package to follow the methodology of Gardner (2022). The package does not report R^2 or adjusted R^2 .

³⁸ In the summer of 2006, lawmakers in Massachusetts increased the minimum wage from \$6.75 to \$7.50, effective January 1, 2007.

³⁹ See <https://casetext.com/case/amerio-v-townsend-oil-co>, <https://www.llrllaw.com/wp-content/uploads/2015/10/Mass-Layers-Weekly.pdf>, and <https://www.lexology.com/library/detail.aspx?g=2995abf0-8e91-45c2-9812-e153fad402aa> (accessed March 23, 2023) for more detailed descriptions.

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